

Final Project Report

Project: Sports HUB Management System Group: BBICT Group Four

Group members

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1. Introduction

A sports club needed a system to manage members and payments online instead of using paper and spreadsheets. We built a web application to do this.

2. Objectives

- Register customers and instructors.
- Let an admin approve members.
- Manage membership categories and subscriptions.
- Handle referrals, reward points and withdrawals.
- Control facility access and produce reports.

3. What we built

A full web app with:

- A **React** frontend (pages for login, dashboard, payments, facilities, rewards, referrals, admin, reports).
- An **Express** REST API with role-based access.
- A **PostgreSQL** database with 12 tables.

All 12 modules in the assignment were implemented (registration, approval, categories, referrals, payments, status, facilities, rewards, withdrawals, transactions, reports, notifications).

4. Tools

React (Vite), Node.js + Express, PostgreSQL, JWT, bcrypt, Git/GitHub, Vercel (hosting) and Render (database).

5. How we worked

We used Git with a `develop` branch and one branch + pull request per task, testing each feature before merging.

6. Challenges

- Getting subscription expiry dates to calculate correctly.
- Fixing facility access (the word "Football" accidentally matched "all").
- Setting up deployment on Vercel with the database on Render.

We solved each of these and tested again.

7. Conclusion

The system works and meets the requirements. In the future we could add email notifications, online payment, and automated tests.